

TandemSDR v1.2

TandemSDR is a self-contained SDR remote control that runs entirely in a browser with no installation required. It links to SDRplay's SDRconnect software via WebSocket and provides a full receiver control interface including spectrum display, waterfall, audio output, signal metering (dBm and SNR), and signal history charting. Program is open source and may be modified and redistributed under GPL v3.

Features

- **WebSocket connection to SDRconnect**

- Connects to SDRconnect running locally or on a network.
- Default connection: 127.0.0.1 port 5454 (both configurable). A hostname like *labcomputer* can also be used if SDRconnect is in Windows admin mode.
- *IP addresses and names are saved in browser local storage and survive page reloads. Accessible from dropdown menu. Backup and restore to json file.
- Will connect over the WAN if router has port 5454 forwarding enabled.
- Connection status indicator shows Connected, Connecting, or Disconnected.
- On connect, all current device state is retrieved automatically from SDRconnect.

- **Frequency Control**

- Separate VFO and Center frequency inputs in MHz.
- Hold-to-repeat step buttons and mouse scroll wheel for VFO tuning. Step buttons accelerate the longer they are held.
- Selectable step size: 10 Hz, 100 Hz, 500 Hz, 1 kHz, 2.5 kHz, 5 kHz, 9 kHz, 10 kHz, 12.5 kHz, 25 kHz, 100 kHz.
- Hold-to-repeat step buttons for Center frequency using the selected step size.
- VFO → Center button to sync the center to the current VFO frequency. When sample rate is above 2 MSPS a 10 kHz offset is applied to keep the VFO out of the DC centre of the spectrum.

- **Device Controls**

- Sample rate selector: 62.5 kSPS to 10 MSPS.
- Device selector — populated dynamically from SDRconnect.
- Antenna selector — populated dynamically from SDRconnect.
- RF Gain slider with overload warning indicator.

- **Demodulation**

- Modes: AM, SAM (Synchronous AM), USB, LSB, CW, NFM, WFM.
- Each mode sets a default filter bandwidth on selection.
- Filter bandwidth input with hold-to-repeat step buttons that accelerate the longer they are held. Maximum bandwidth is enforced per mode.
- WFM mode requires a minimum sample rate of 250 kSPS; the 62.5 and 125 kSPS options are disabled when WFM is selected. If either of those rates is active when WFM is selected, the sample rate automatically jumps to 2 MSPS.
- RDS on/off toggle with PI, PTY, and PS display (WFM mode).
- Stereo on/off toggle with stereo indicator.

- **Audio**

- Audio in SDRconnect is muted after TandemSDR is connected.
- Audio streamed from SDRconnect as packets and decoded in the browser via the Web Audio API.
- Volume control slider with mute toggle button.
- Noise Reduction on/off toggle with strength slider.
- Audio AGC on/off toggle with threshold slider.
- Squelch on/off toggle with level slider.

- **S-Meter**

- Segmented bar display calibrated to IARU standard.
- HF mode (below 30 MHz): S9 = -73 dBm, 6 dB per S-unit.
- VHF mode (30 MHz and above): S9 = -93 dBm, 6 dB per S-unit.
- Segments S1–S9 in green, +10 through +60 dB over S9 in red.
- Automatically switches HF/VHF calibration based on current frequency.

- **Signal readouts**

- Live Signal Power display in dBm.
- Live Signal SNR display in dB.
- Live Frequency display.

- **Spectrum Display**

- FFT spectrum analyzer. (512 points)
- Adjustable Max (reference) and Min (base) level inputs.

- Frame averaging: None, 2, 4, 8, 16 frames.
- Zoom: $\times 1$, $\times 2$, $\times 4$ — cycled via button or keyboard.
- Frequency snap grid: 1 kHz, 5 kHz, 9 kHz, 10 kHz, 25 kHz, 100 kHz.
- Crosshair follows mouse with frequency readout.
- Left-click sets VFO frequency at cursor position.
- Right-click sets Center frequency at cursor and syncs VFO.
- Drag on the frequency label row to pan the center frequency.
- Scroll wheel steps the VFO frequency.
- **Waterfall Display**
 - Scrolling waterfall below the spectrum.
 - Optional frame averaging (follows spectrum averaging setting).
 - Speed selector: Slow, Medium, Fast.
 - Same mouse controls as the spectrum (left-click, right-click, scroll wheel).
- **Signal History Chart**
 - Time-series chart of Signal Power (dBm) or SNR over time.
 - Selectable capture interval: 500 ms, 1 sec, 5 sec, 10 sec, 30 sec, 60 sec.
 - Start/Stop/Clear capture control.
 - Session min and max values with timestamps displayed on the chart.
 - Scrollable history slider to review up to $10\times$ the visible window of captured data. Automatically snaps back to live after 20 seconds of inactivity.
 - Crosshair mouseover shows exact value and timestamp when stopped or in history mode.
 - **Record** — opens a native file save dialog and streams data directly to a CSV file in real time as samples are captured.
 - **Export** — saves the current in-memory buffer to a CSV file via a native file save dialog. CSV includes frequency, mode, capture interval, and timestamped power and SNR columns.
- **Memory**
 - 10 memory slots for saving and recalling receiver settings.
 - Each slot stores VFO frequency, center frequency, mode, filter bandwidth, sample rate, antenna, RF gain, step size, volume, and spectrum display settings (reference level, base level, zoom, averaging, snap).

- Slots are saved in browser local storage and survive page reloads. Backup and restore buttons save memories in a json file.
- **Band Selector**
 - Quick-access modal for jumping to common bands with one click.
 - Sets center frequency, sample rate, mode, snap, and step for the selected band. Also sets RF gain to maximum and turns squelch off.
 - Covers Amateur Radio (160 m through 33 cm), Shortwave Broadcast (120 m through 11 m), and Broadcast (AM and FM).
- **Keyboard Shortcuts** (hotkeys)

Key	Action
← →	Step VFO by selected step size
↑ ↓	Step Center by Snap value
PgUp PgDn	Jump Center one full span width
=	Copy VFO → Center
A	AM mode
S	SAM mode
U	USB mode
L	LSB mode
C	CW mode
N	NFM mode
W	WFM mode
V	Toggle audio mute
Q	Toggle squelch
D	Toggle Spectrum Display
F	Toggle WaterFall (enables Spectrum first if hidden)
Z	Cycle spectrum zoom (×1 → ×2 → ×4)
+	Increase sample rate
-	Decrease sample rate
M	Save memory slot
R	Recall memory slot
B	Open Band Selector
H or ?	Open Help
Esc	Close any open modal

- **Built-in Help**

- Help modal accessible via the ? button or H/? keyboard shortcut.
- Covers all mouse controls for spectrum and waterfall.
- Full keyboard shortcut reference.

Requirements

- No installation and no internet connection required. Open the HTML file directly in your browser
- **SDRplay SDRconnect** must be running with WebSocket access enabled on port 5454
- Compatible with any SDRplay RSP device supported by SDRconnect.
- Works in any modern browser. The **Record** and **Export** features require a Chromium-based browser (Google Chrome, Microsoft Edge, Brave, or Opera) as they depend on the File System Access API, which is not supported in Firefox or Safari.

Typical Workflow

1. Start **SDRconnect** and ensure it is running with WebSocket enabled. See image below
2. Open **TandemSDR** by double-clicking the HTML file in your browser.
3. Enter the IP address and port of the SDRconnect instance (defaults to 127.0.0.1:5454).
4. Click **Connect**. TandemSDR will retrieve all current device settings automatically.
5. Use the **VFO** input or click on the **Spectrum** to tune to a frequency.
6. Select a demodulation **Mode** and adjust **Filter BW** as needed.
7. Adjust **RF Gain**, **Volume**, **Noise Reduction**, and **Squelch** as desired.
8. Click **Show** under Spectrum Display to enable the spectrum and waterfall.
9. Use **Signal History** to capture and export signal power or SNR over time.
10. Hotkeys (listed above) allow quick changes to features and parameters.

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**** Enable WebSocket in SDRconnect before using program. **